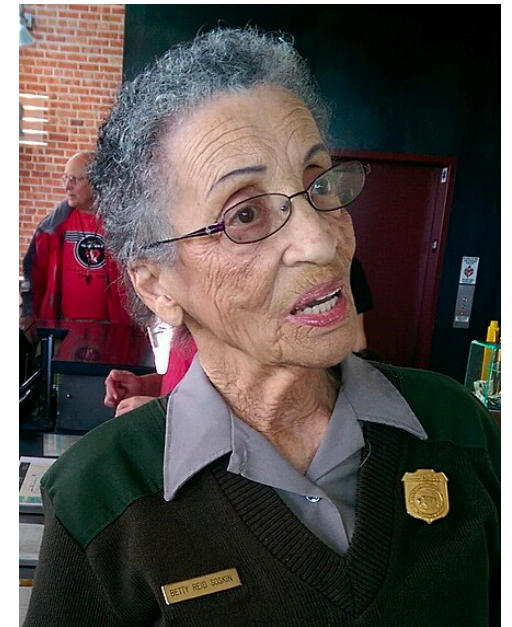


[Wendy the Welder in the
Kaiser Richmond Shipyards](#)



[Betty Reid Soskin \(1921-2025\)](#)
National Park Ranger until age 100

ECON 100B: Macroeconomics

Class 12: The Labor Market (Jones Chapter 7)

Ryan D. Edwards
February 26, 2026

[Picture of children at a protest in
Columbus, Ohio, around 1929](#)



Agenda

- Announcements
- Current events
- Jones Chapter 7: The Labor Market
 - Will be covered on the final exam
 - NOT on the midterm exam



BERKELEY STUDENT LEARNING CENTER

ECONOMICS PROGRAM

Mastery through Collaborative Learning

ECON 100B

César Chávez Student Center

Monday - Thursday 12 - 5 PM



Service Schedule

Announcements

- Problem Set 2 is due on Gradescope this Friday 2/27
- In-class REVIEW next Tuesday 3/3
- In-class Midterm next Thursday 3/5
- Take-home Midterm Essay released 3/8, due 3/20
- Spring Break 3/23-3/27

Midterm exam

- DSP students
 - If you have more than 150% time, please set up DSP proctoring through the DSP portal
 - If you have 150% time only, then we will invite you to a classroom in Giannini Hall. Plan to start **promptly at 9:30am** to get 90 minutes. The rooms will be empty; they are reserved starting at 9:00am
- The main class reports to Pimentel Auditorium for a 60 minute exam. We will start at 9:40am or slightly later, end at 10:40am+
 - **Sit where you like.** Note and write down your seat number and row
- Everyone gets 1 page front+back of notes

Cheating is rare. But we take it seriously

- In Spring 2025, a student sat in the back row in Pimentel Auditorium and copied off their neighbors
- The student also appeared to use a cell phone
- The neighbors noticed this and requested reseating. We did that, and we helped them. They did not have to testify
- The affair was reported to the Center for Student Conduct, which has a hearing pending still
- We will do our jobs, no matter how long. Cheating brings consequences

Current events

- I miss the Olympics!

Learning objectives in Chapter 7 (1/2)

- Labor demand slopes down and labor supply slopes up
- The 20th Century saw large increases in female labor supply
- In the U.S., **unemployment** rises a lot and falls a lot. Other countries do it differently, subsidizing firms not to lay off workers. Unemployment includes:
 - **Frictional**: Jobs and businesses often start and end, and workers reallocate
 - **Structural**: Workers' skills might be mismatched with firms' needs
 - **Cyclical**: During business cycle troughs, labor demand usually falls
 - **Seasonal**: Weather and holiday seasons affect labor demand
- *Involuntary unemployment* is what feels socially unacceptable
- Some policies exacerbate unemployment. See ECON 151

Learning objectives in Chapter 7 (2/2)

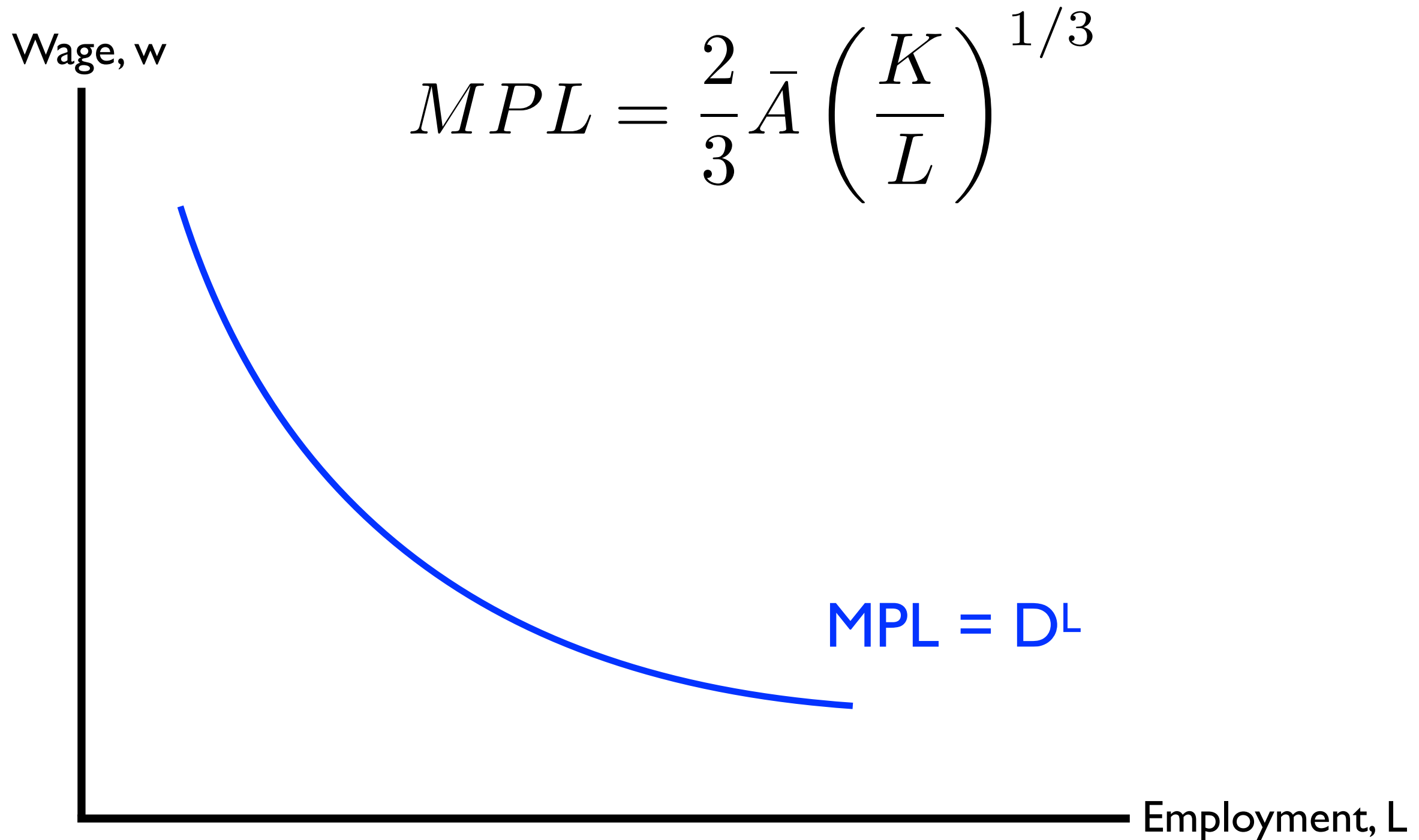
- The **present discounted value** (PDV) of earnings is very large.
With an annual real interest rate $R = 0.03$

$$PDV = \sum_0^{44} \frac{100,000}{(1 + 0.03)^t} \approx 2.53 \times 10^6$$

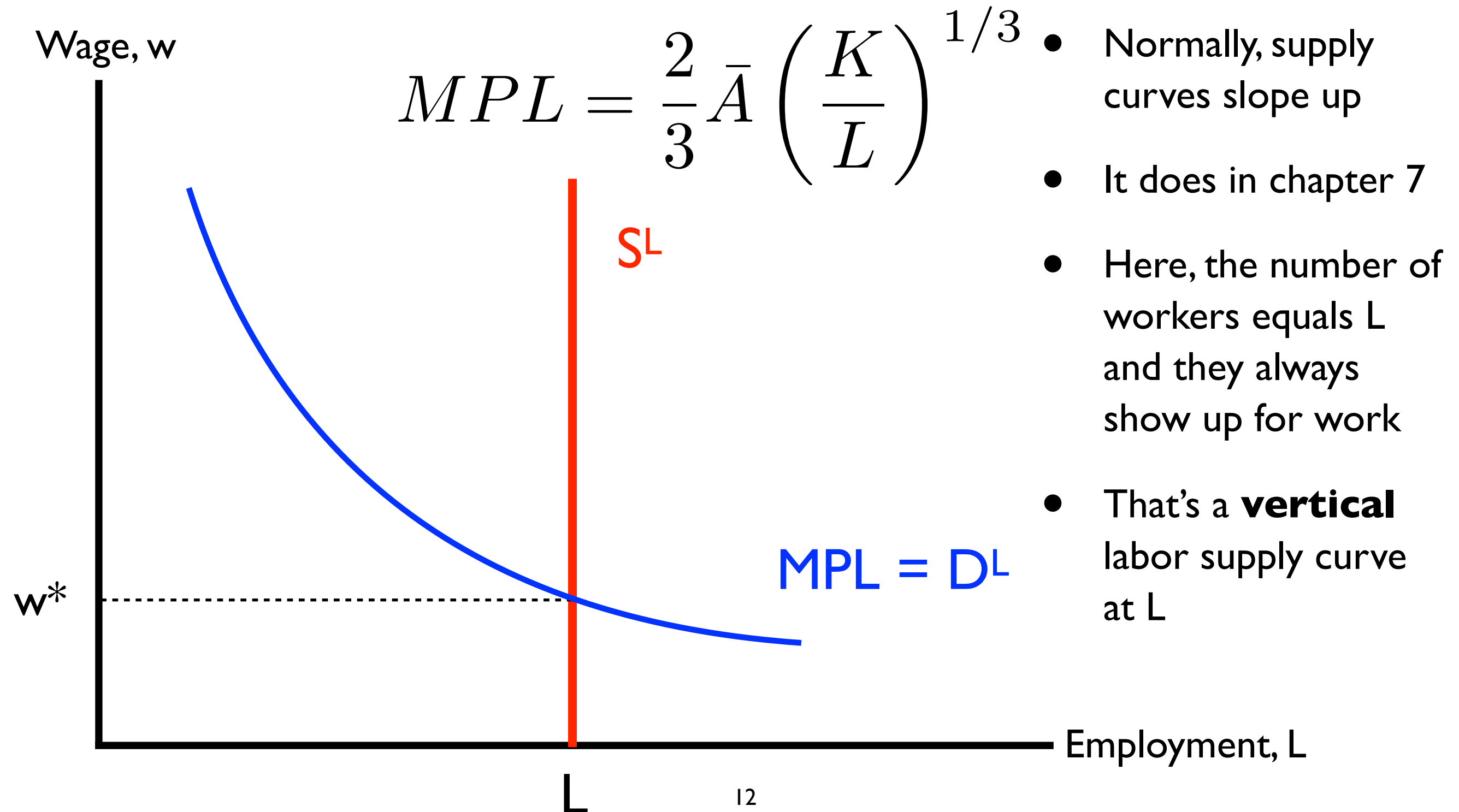
- The **college wage premium** is large and has grown, even through expansions of labor supply by college educated workers
- For much, much more, see ECON 151: Labor Economics

**Recall the simple labor market
inside the neoclassical model in
ECON 100B**

A factor's marginal product is its **demand curve**



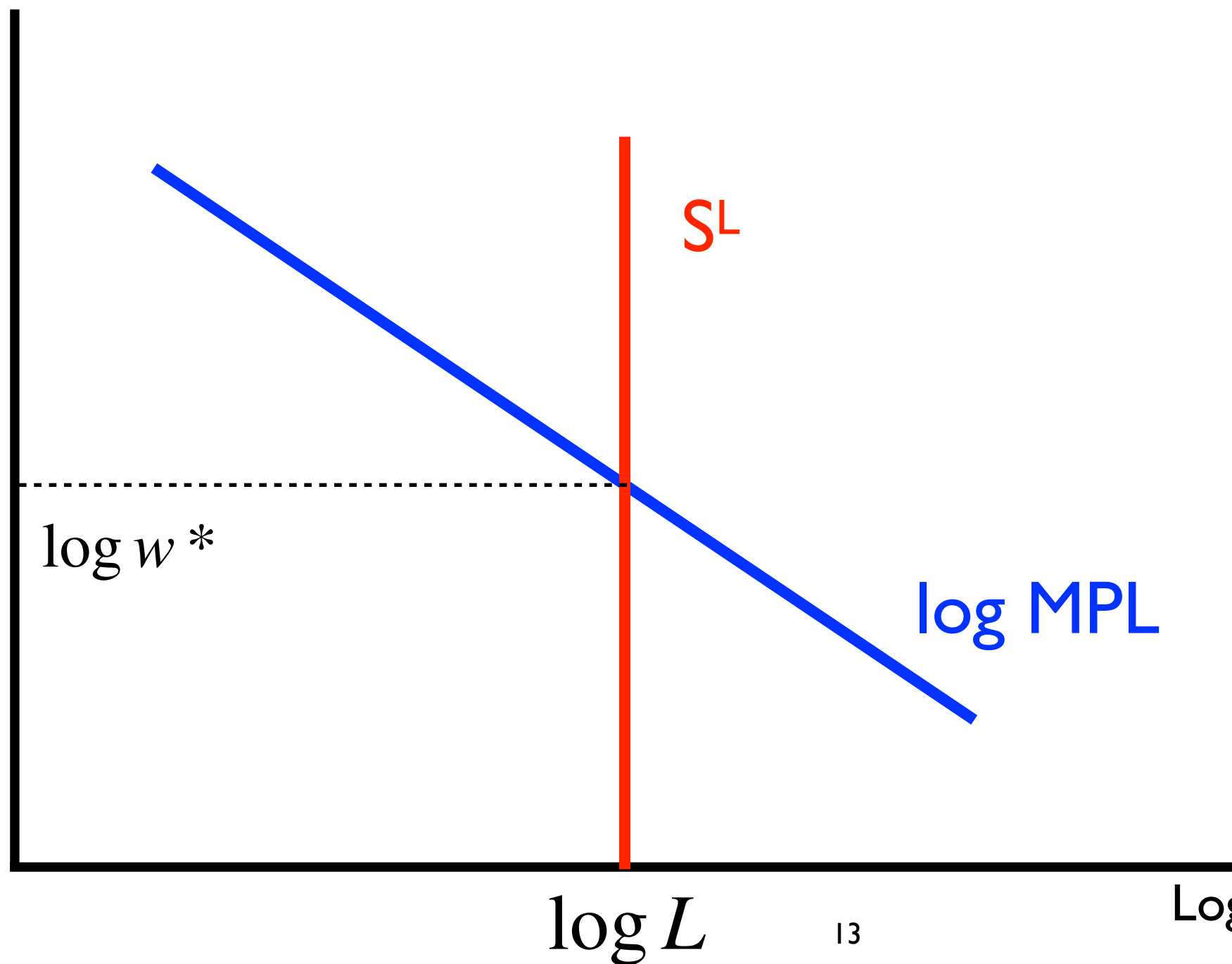
In Chapter 4, labor supply is simply the quantity of people



Suppose we took logs of both sides

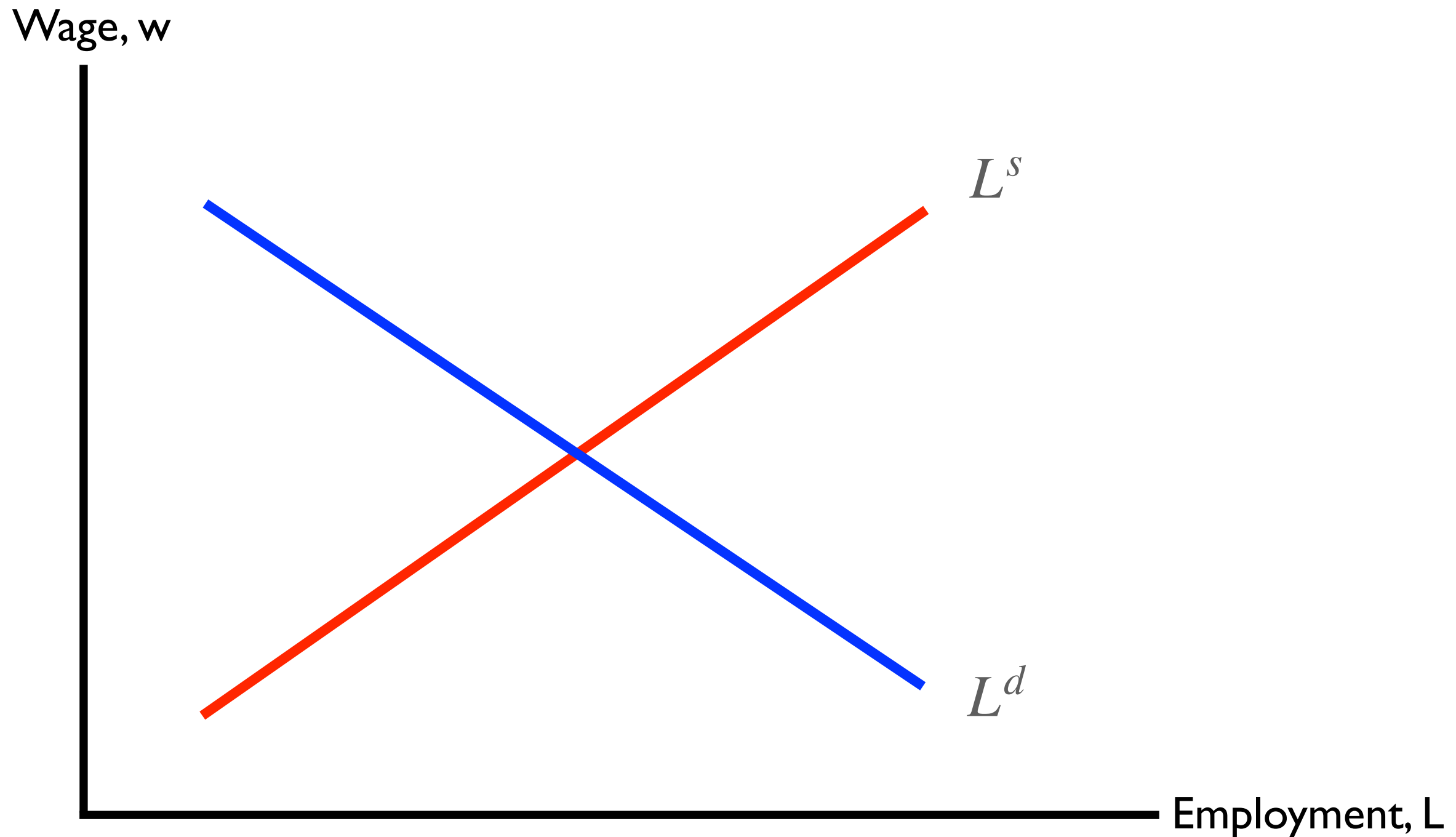
$$\log w = \log \left(\frac{2}{3} \right) + \log \bar{A} + \frac{1}{3} \log K - \frac{1}{3} \log L$$

Log wage, $\log w$

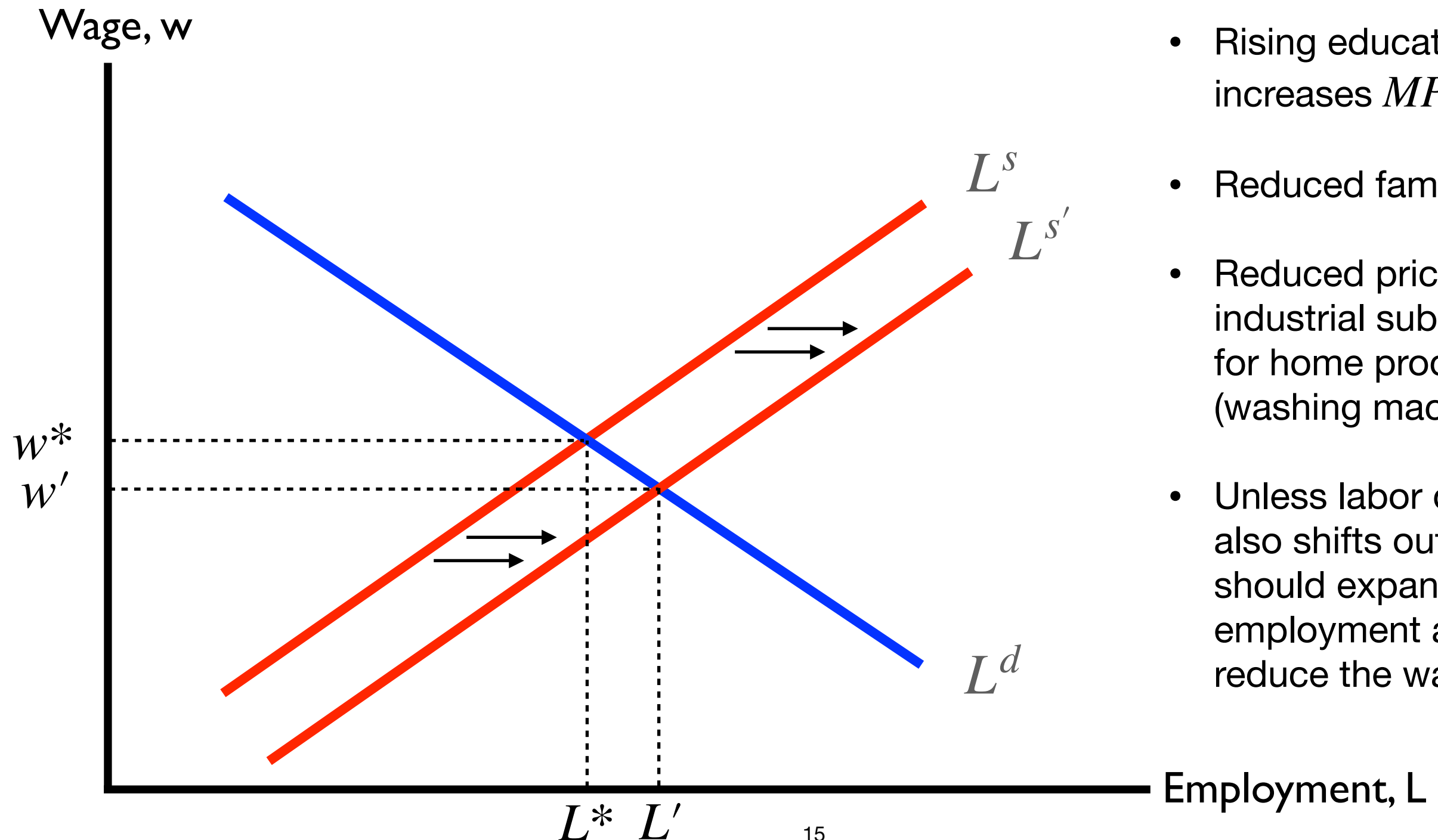


- The log of a product is the sum of the logs
- What was convex is now linear

In Chapter 7, let's assume linear supply and demand curves for labor



In advanced economies during the 20th century, female labor supply expanded

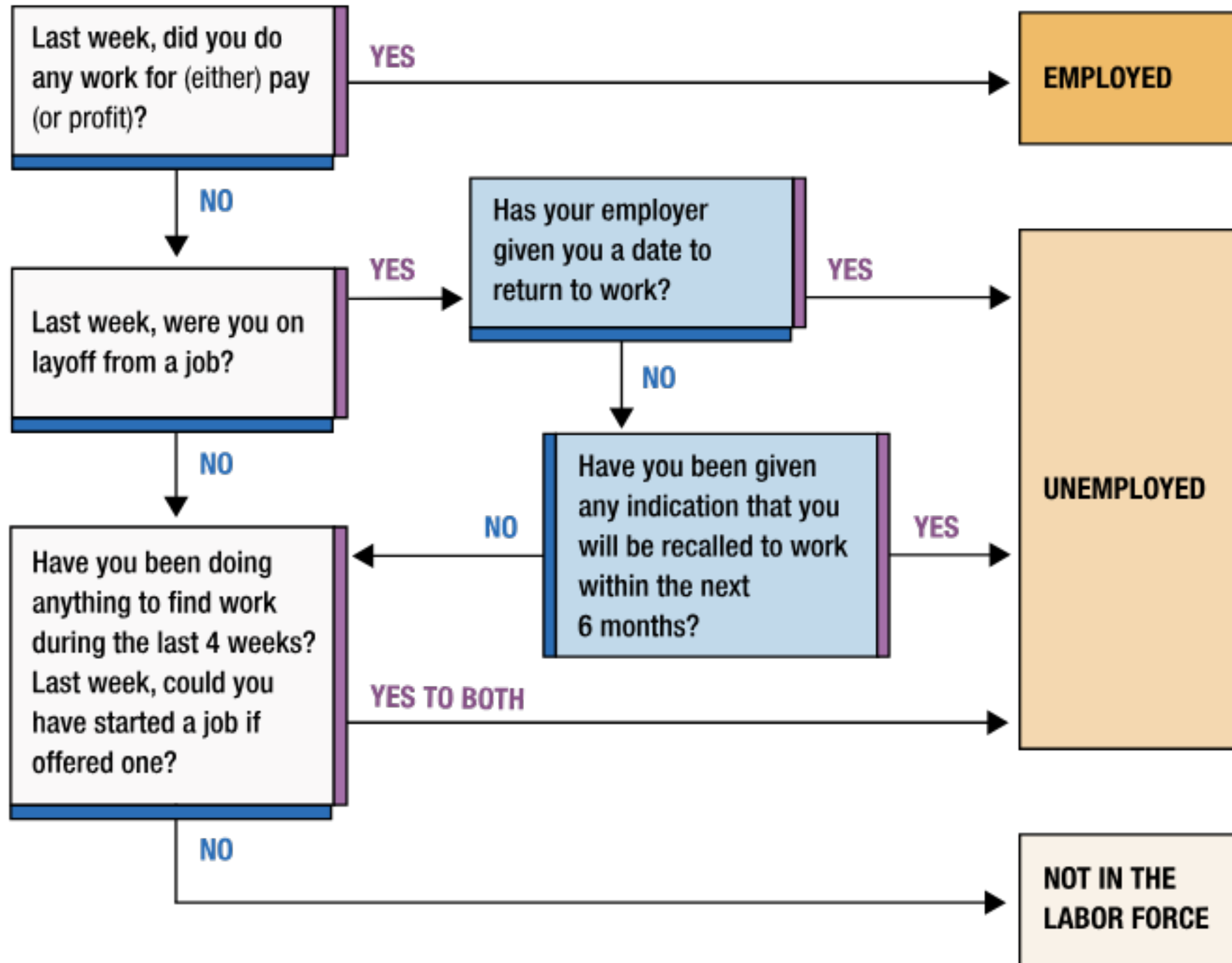


- Economic explanations:
- Rising education increases MPL
- Reduced family size
- Reduced prices of industrial substitutes for home production (washing machines)
- Unless labor demand also shifts out, this should expand employment and reduce the wage

How do we measure?

- The Bureau of Labor Statistics has conducted the monthly **Current Population Survey** in a similar format since the late 1940s
- (Pro tip: Check out <https://cps.ipums.org> for data)
- Monthly [CPS Methodology](#):
 - Outreach via telephone, ends with about 60,000 households and maybe 110,000 people
 - Occurs during the week including the 19th day, asks about the previous week back to the 12th day
 - Asks a bunch of questions. For employment, the key sequence appears on the next slide

From RAND, an interpretation of the CPS questionnaire



Clever measures and variable names

- All are measured for the civilian noninstitutionalized population aged 16+

- Employed (ages 16+) L

- Unemployed (ages 16+) E

- Labor force (ages 16+) $LF = E + U$

- Population (age 16+) N

- Employment-to-population ratio $empop = \frac{E}{N}$

- Labor force participation rate $LFPR = \frac{E + U}{N}$

Some measures wiggle more than others

- Business cycles are recessions and expansions. During recessions, unemployment U rises and employment E falls

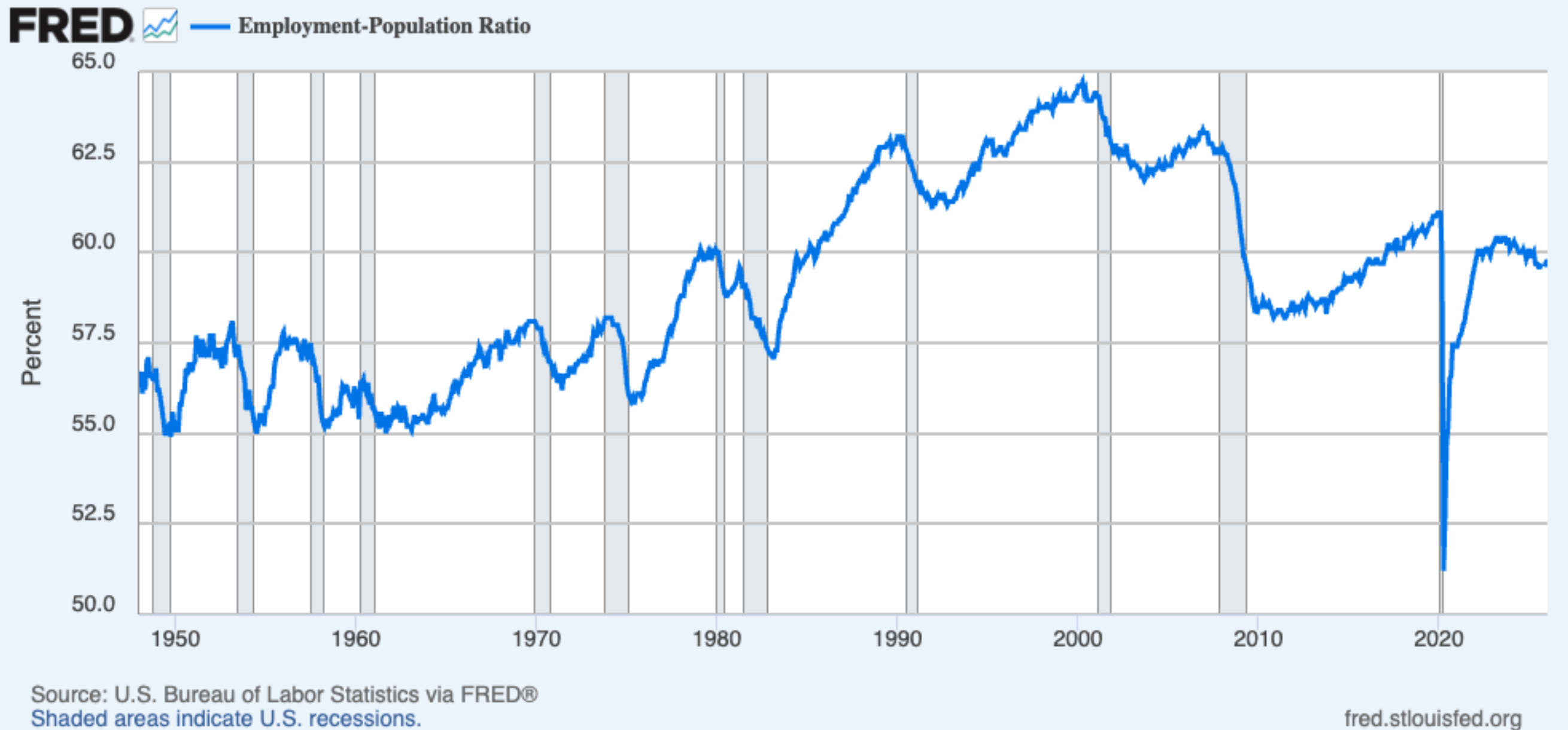
- Employment-to-population ratio $empop = \frac{E}{N}$

- Labor force participation rate $LFPR = \frac{E + U}{N}$

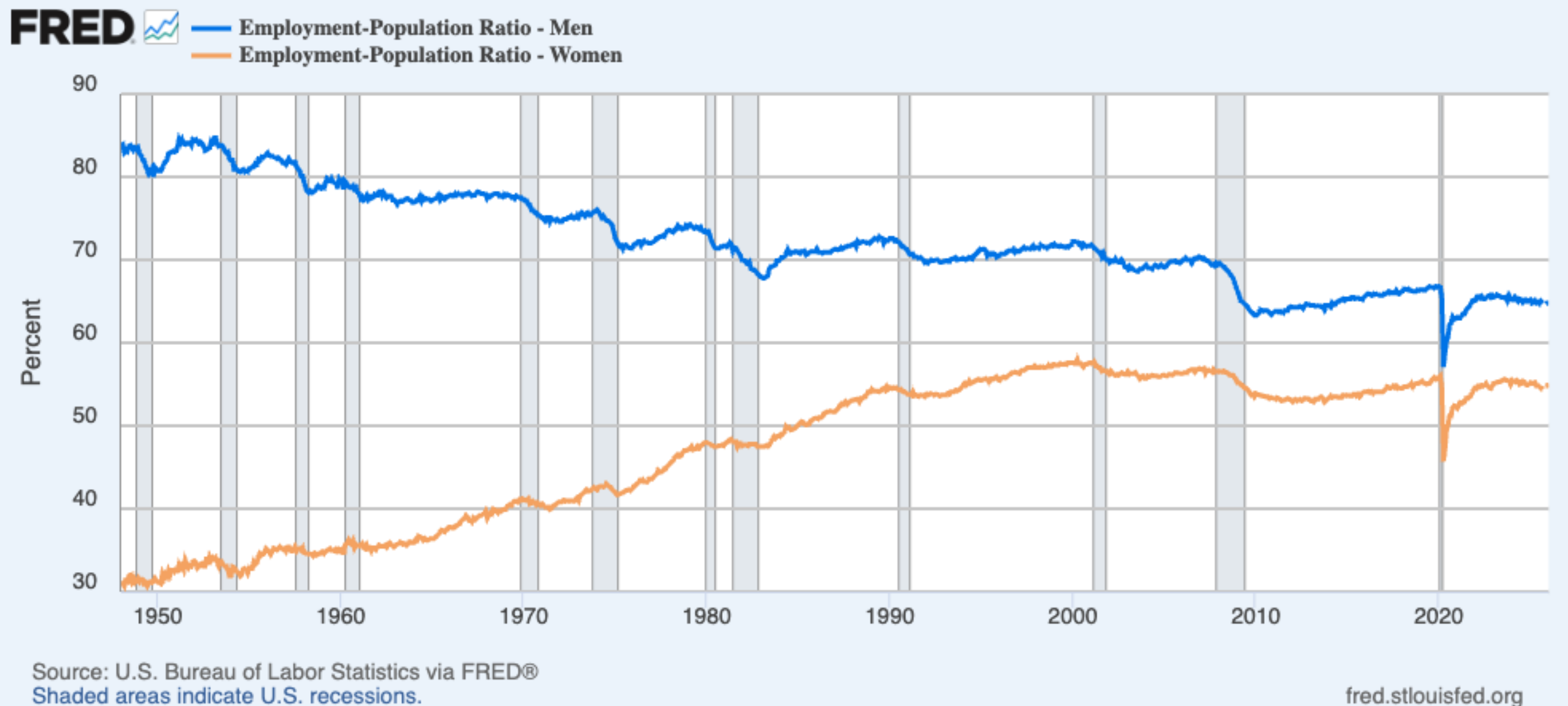
- Shifts from E into U and back will not affect $LFPR$ because the numerator remains the same

but they will affect $empop$

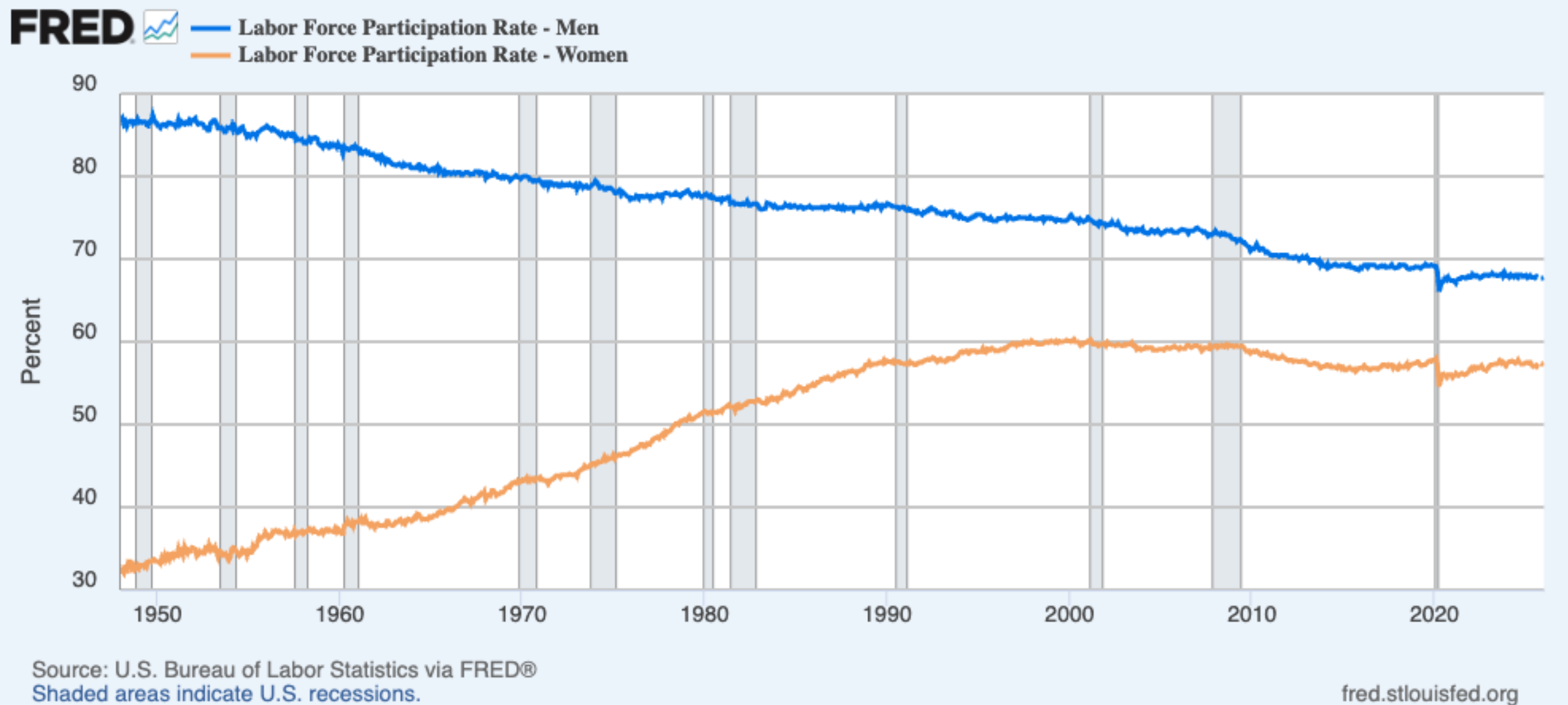
empop via FRED for males and females combined shows increases starting in maybe the 1960s and lots of wiggles



empop for males has drifted lower, while for females it had risen up until the 2000s



LFPRs show the long-run trends more clearly

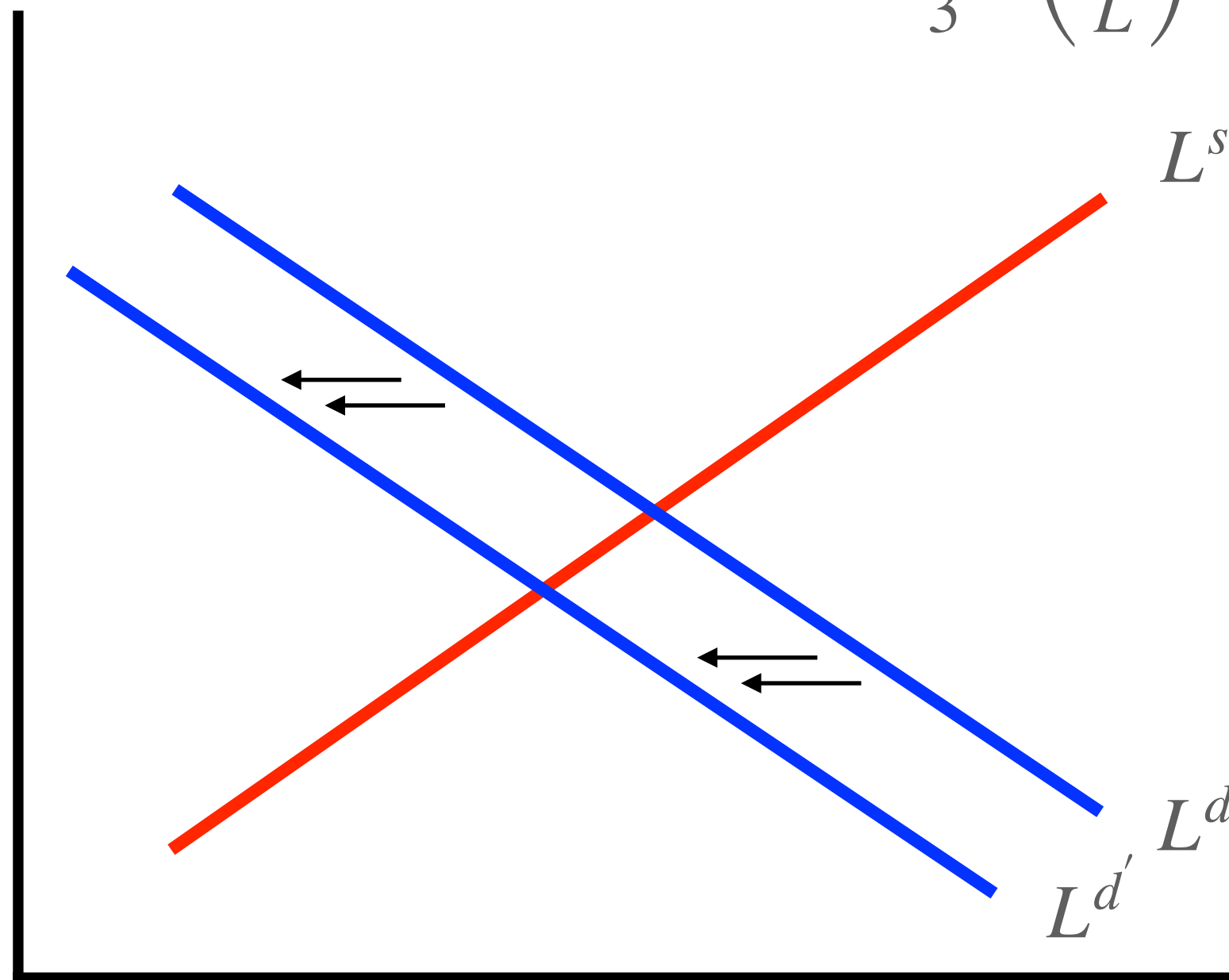


Labor demand can also shift. During recessions, it usually shifts inward

$$L^d = MRPL = MR \cdot MPL$$

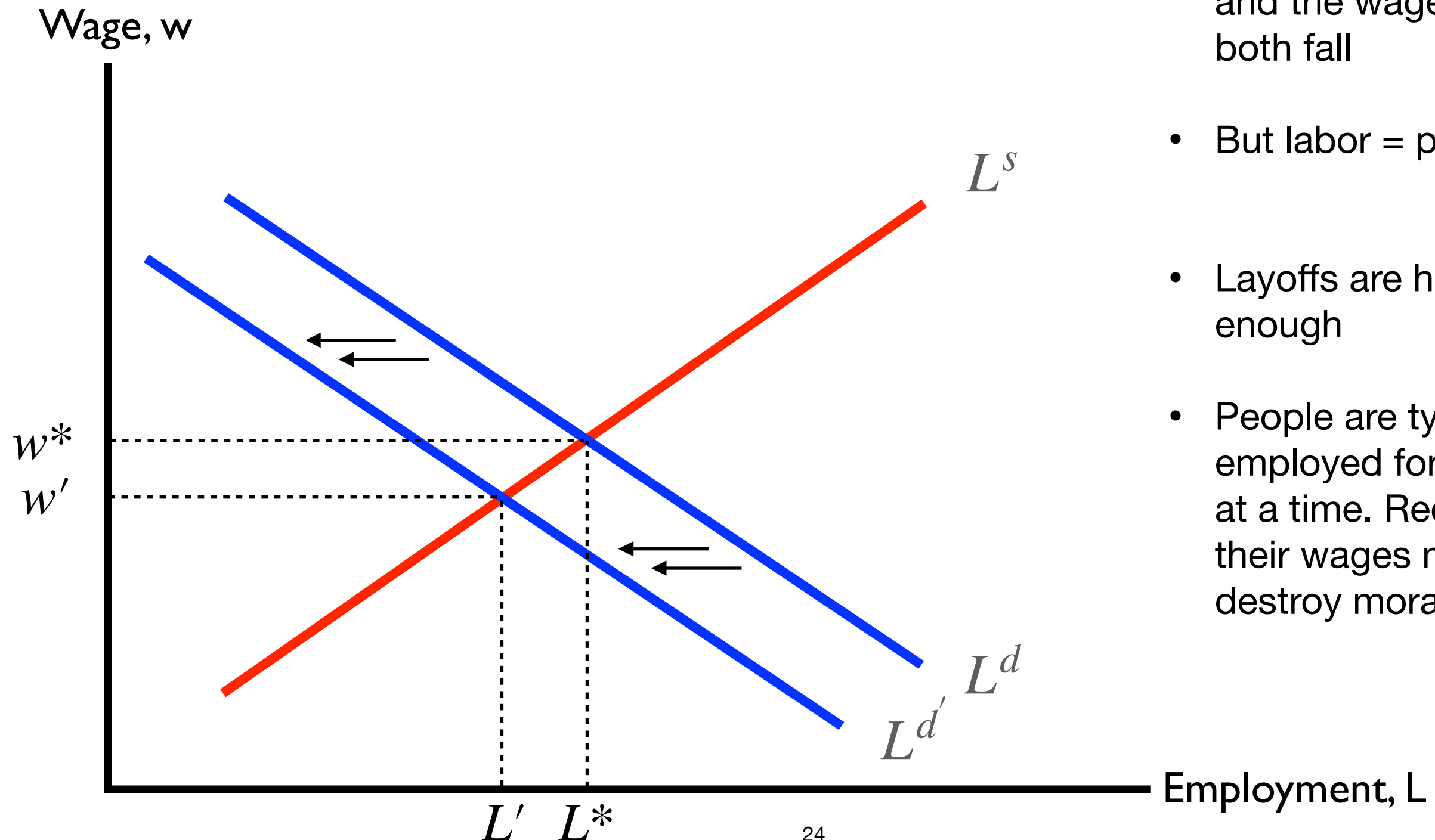
$$= MR \cdot \frac{2}{3} A \left(\frac{K}{L} \right)^{1/3}$$

Wage, w



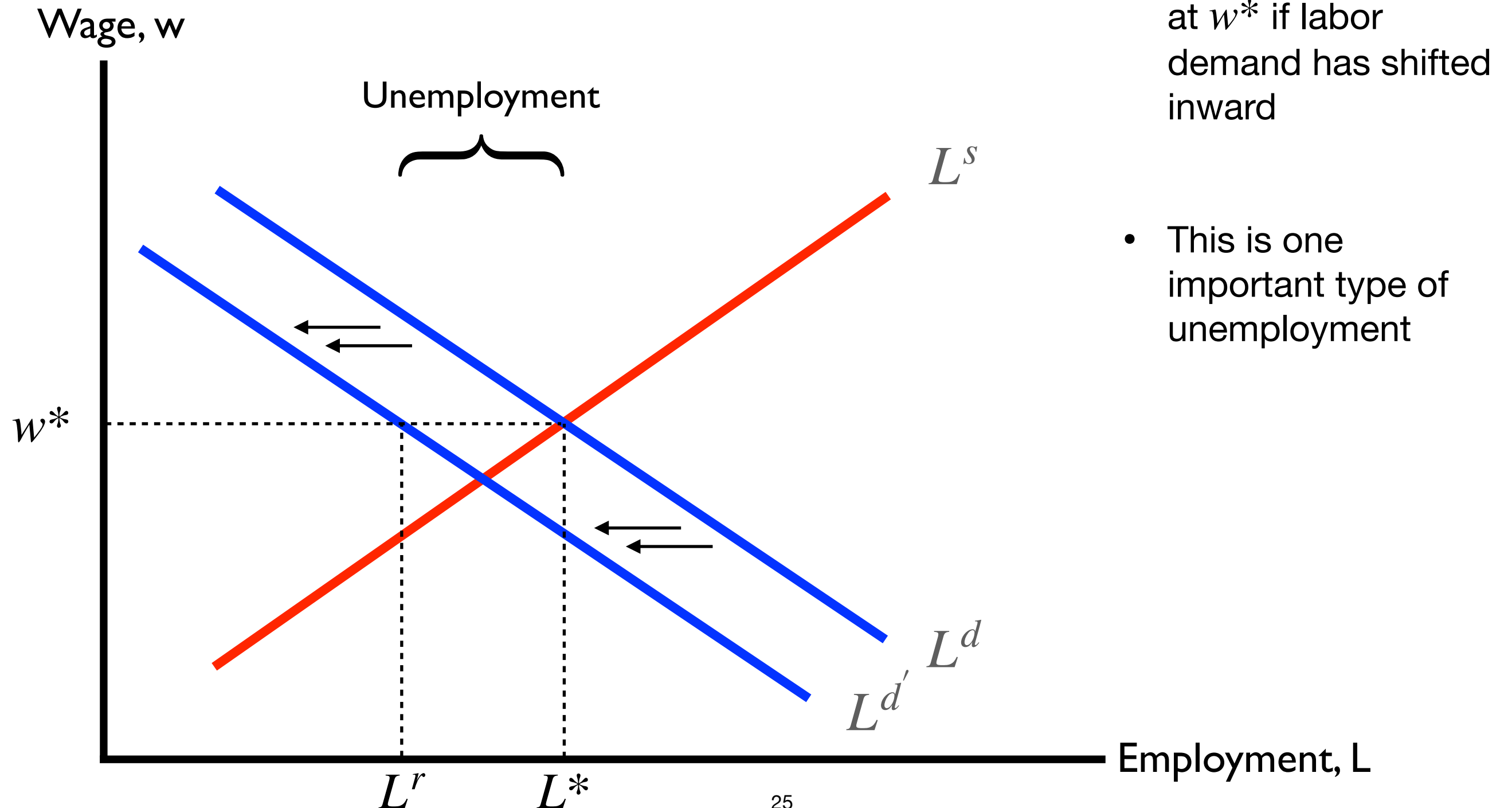
- You can think of this as due to a cyclical reduction in either
 - Total factor productivity A and thus in MPL
 - The demand for the product and thus in marginal revenue MR , also known as the market price of the product

In a recession, a perfect market sheds and reprices labor



- For the labor market to clear, employment and the wage should both fall
- But labor = people
- Layoffs are hard enough
- People are typically employed for years at a time. Reducing their wages might destroy morale

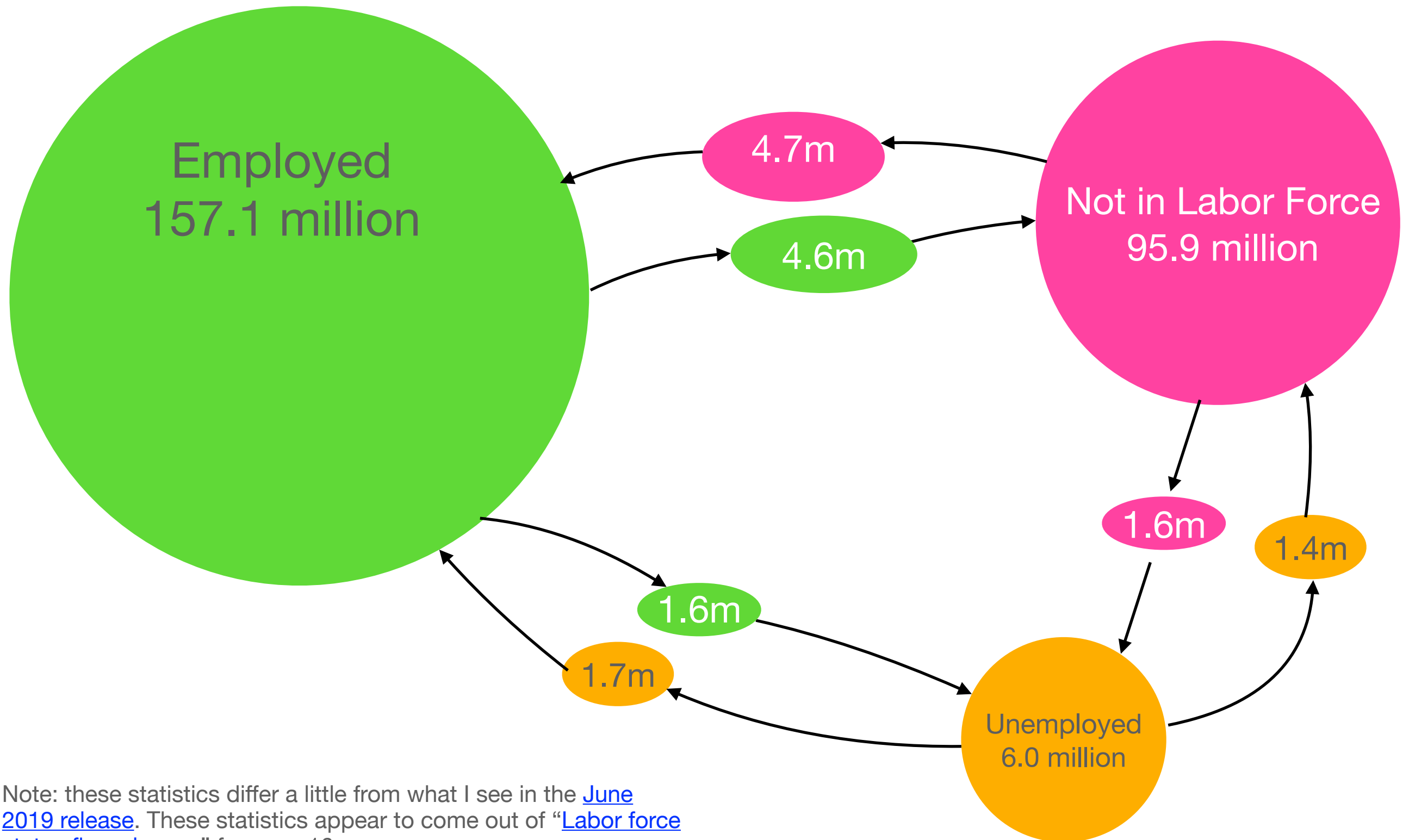
Involuntary cyclical unemployment results if the market wage fails to adjust downward when labor demand recedes



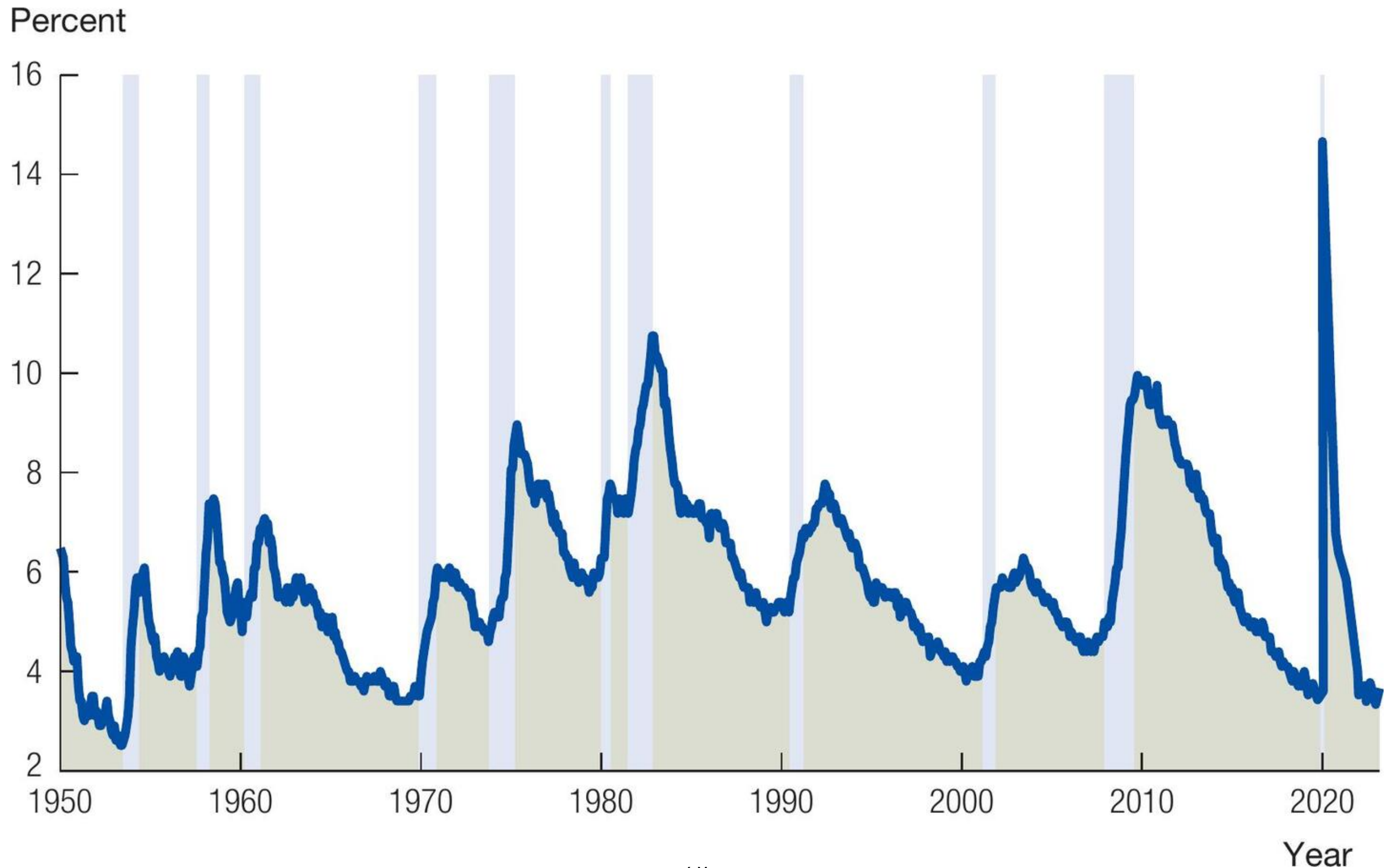
Types of unemployment

- **Frictional:** Jobs and businesses start and end, workers move on
 - **Structural:** Workers' skills might not match firms' needs
 - **Cyclical:** During recessions, labor demand usually falls
 - **Seasonal:** Weather and holiday seasons affect labor demand
-
- Frictions mean that even a healthy labor market will still include some temporarily unemployed workers who will move on
 - The nasty one here is structural

The labor market May-June 2019



The unemployment rate varies strongly in the U.S. over the business cycle (shown here in light blue vertical shading)



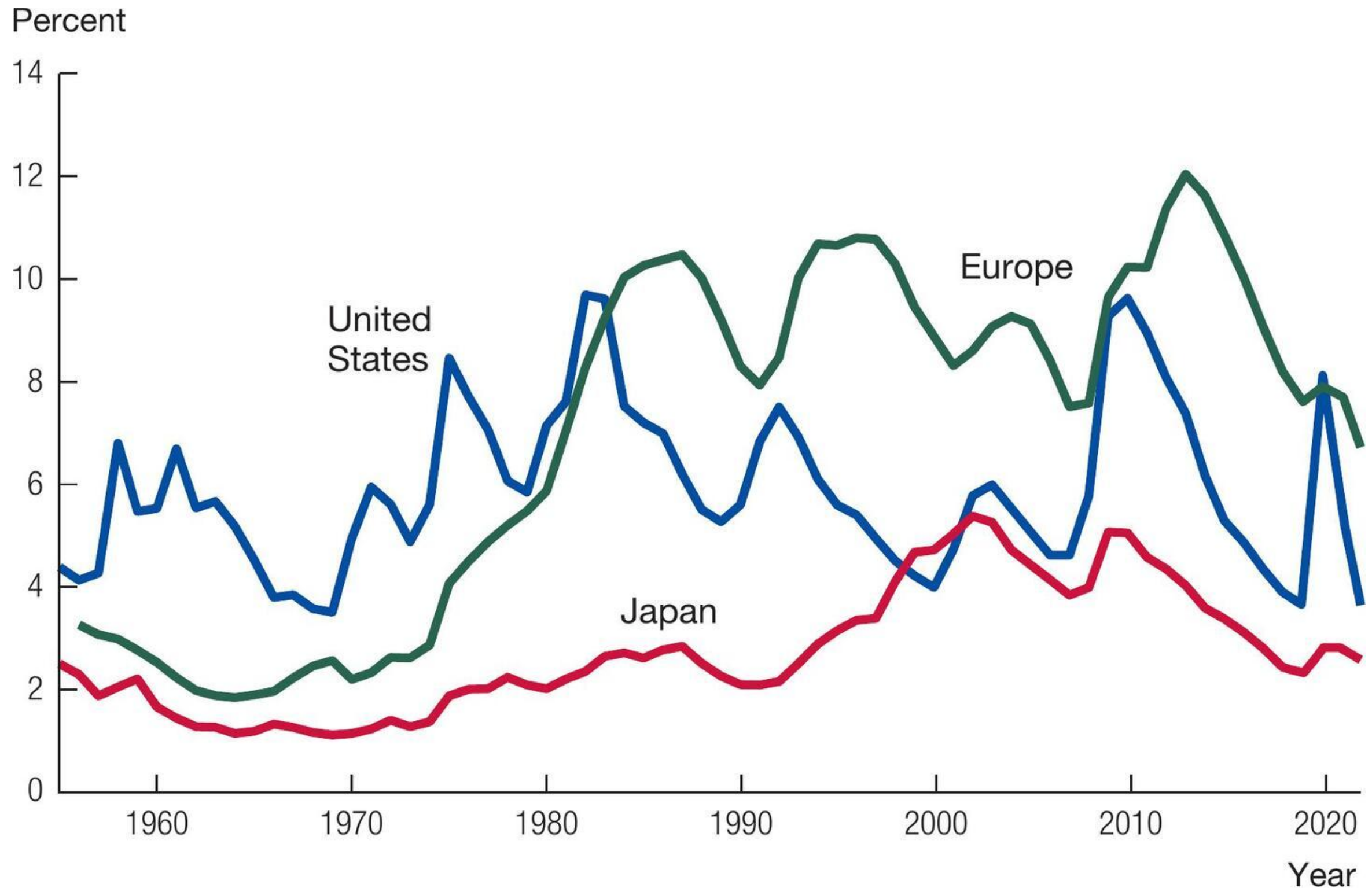
OECD countries vary in terms of unemployment rates, reasons, and identities. From the e151 textbook [ESH 14e](#)

Table 14.3 Unemployment and Long-Term Unemployment, Selected European and North American Countries, 2005 and 2019

	Unemployment overall rate (%)		Percent of unemployed out of work > one year	
	2005	2019	2005	2019
Belgium	8.4	5.4	51.6	43.5
Canada	6.8	5.6	9.6	8.5
Denmark	4.8	4.7	25.9	16.6
France	9.5	8.5	42.5	38.8
Germany	9.5	3.1	54.0	38.1
Ireland	4.3	5.1	34.3	33.3
Netherlands	4.8	3.4	40.1	31.4
Norway	4.6	3.6	9.4	24.1
United Kingdom	4.7	3.8	22.4	25.0
United States	5.1	3.7	11.8	12.7

Sources: OECD, *Employment Outlook* (Paris: OECD, 2006), Tables A and G; OECD Data, Unemployment Rate at <https://data.oecd.org/unemp/unemployment-rate.htm>; and Long-Term Unemployment Rate at <https://data.oecd.org/unemp/long-term-unemployment-rate.htm>

Fluctuations in the unemployment rate across the business cycle are much larger in the U.S. and more muted in Europe and Japan



Source: OECD Main Economic Indicators and the U.S. Bureau of Labor Statistics.

**For more about unemployment,
enroll in ECON 151**

Present discounted value

NY SPORTS

There's No Accounting for This

Thanks to a Deferred Buyout, the Mets Still Owe Bobby Bonilla 25 Annual Payments of \$1.19 Million

 Email

 Print

 Save





15 Comments













By MIKE SIELSKI

Updated July 1, 2010 12:01 a.m. ET

(See *Correction & Amplification* [below](#).)

SAN JUAN, P.R.—One year from today, the Mets will add to their payroll a 47-year-old, past-his-prime power hitter who has a reputation as a malcontent—a player who has been retired from professional baseball for nine years and won't play another game again.

Nevertheless, starting on July 1, 2011, Bobby Bonilla will remain on the franchise's payroll for 25 years, collecting an annual salary of \$1,193,248.20. Those are the terms the Mets agreed to Jan. 3, 2000, when they bought out the final year of Mr. Bonilla's contract.

"That beautiful thing," he said here Monday.



[Enlarge Image](#)

Former Met Bobby Bonilla calls his Mets contract, which will earn him \$29,831,205 between 2011 and 2035, 'that beautiful thing.' Ezra O. Shaw /Allsport

The facts

- In Spring Training in 2000, Bobby Bonilla was entering the last year of his guaranteed contract
- The Mets had to pay him \$5.9 million that year
- But they didn't really want him around!
- In 1999, he batted .160 in 119 at-bats, on a team that finished 97-66 & lost the NLCS to the Braves (who lost World Series to the Yankees)
- If you're the Mets, what do you do? How much do you have to pay Bobby to get lost?
- One answer: \$5.9 million. But what if you *defer your payments*?

Deferred payment

- How much would the Mets have to pay Bobby Bonilla beginning in 2011 so that they can NOT pay him \$5.9 million today?
- The answer depends on **present discounted value** and the interest rate
- The Mets and Bonilla agreed on an interest rate of 8%
- That means they thought this was true:

$$5.9 = \frac{1.19}{(1 + 0.08)^{11}} + \frac{1.19}{(1 + 0.08)^{12}} + \dots + \frac{1.19}{(1 + 0.08)^{35}}$$

(in 2011)35(in 2012)(in 2035)

Present discounted value

$$5.9 = \frac{1.19}{(1 + 0.08)^{11}} + \frac{1.19}{(1 + 0.08)^{12}} + \dots + \frac{1.19}{(1 + 0.08)^{35}}$$

(in 2011)(in 2012)(in 2035)

- The value of a contract paying \$1.19 million starting in 2011 through 2035 had a *present value* in 2000 of \$5.9 million
- If the interest rate was actually 8%,
- This is a lot like Bobby Bonilla loaning the \$5.9 million to the Mets at an interest rate of 8%
- Nice work if you can get it! That 8% is a pretty high rate of return!

Bottom line

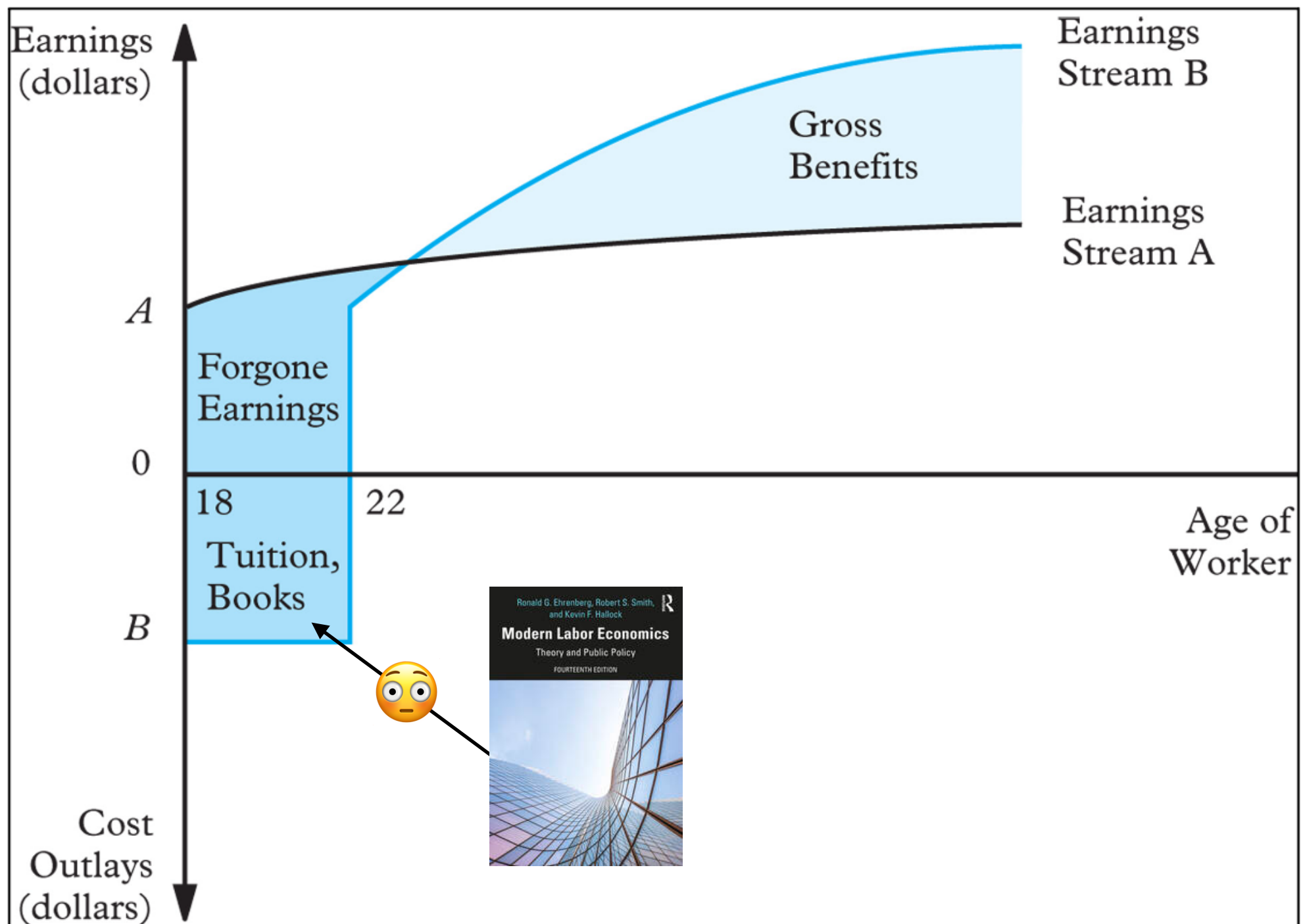
- The Mets entered willingly into an agreement that arguably helped them, at an interest rate close to the prevailing “prime rate”
- They got to the World Series vs. the Yankees in 2000 (“Subway Series”), having signed Todd Zeile as a free agent and traded for Mike Hampton
- They were probably also earning pretty high rates of return from **Bernie Madoff** at the time! Maybe higher than 8%
- The agreement definitely helped Bobby Bonilla, who played for Atlanta in 2000 and St. Louis in 2001
- The actual “prime rate” ended up being lower than 8% for pretty much the entire period, and Bonilla got about 60% more because of it

Learning objective

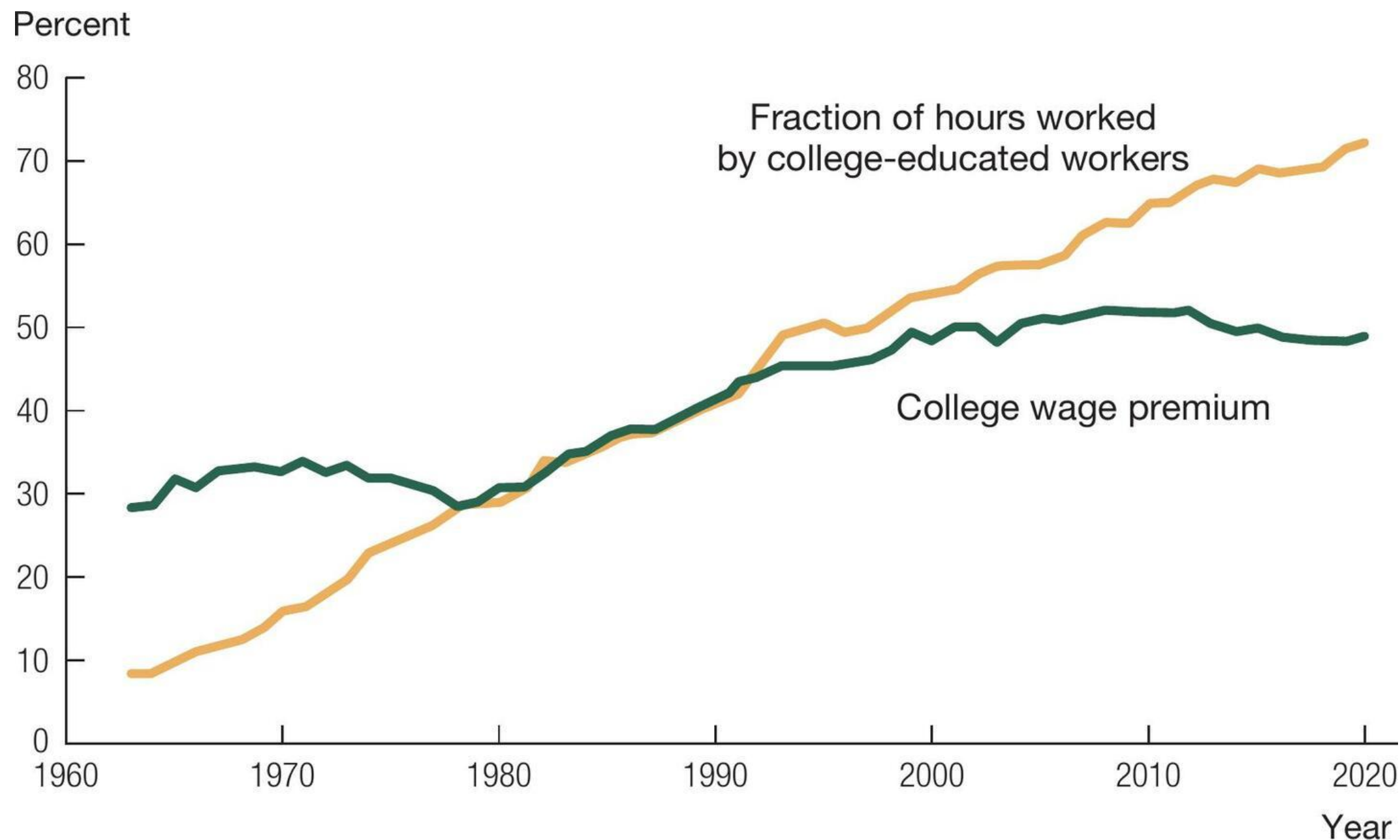
- On an exam, you might see this
- Q: Suppose the Mets paid Bobby Bonilla \$1.19 million a year starting in year 11 through year 35. What is the present discounted value of that contract if the annual interest rate is 8%?
- A:
$$\frac{1.19}{(1 + 0.08)^{11}} + \frac{1.19}{(1 + 0.08)^{12}} + \dots + \frac{1.19}{(1 + 0.08)^{35}}$$
- Variations on this are fine! The key is to be able to recognize and work with the basic format of present discounted value

For any investment with long-lived payouts and up-front costs, like attending college, present value is important. From the e151 textbook [ESH 14e](#)

Figure 9.2 Alternative Earnings Streams



The additional earnings associated with a college degree are big and show no signs of collapsing, even though college-educated labor supply has expanded

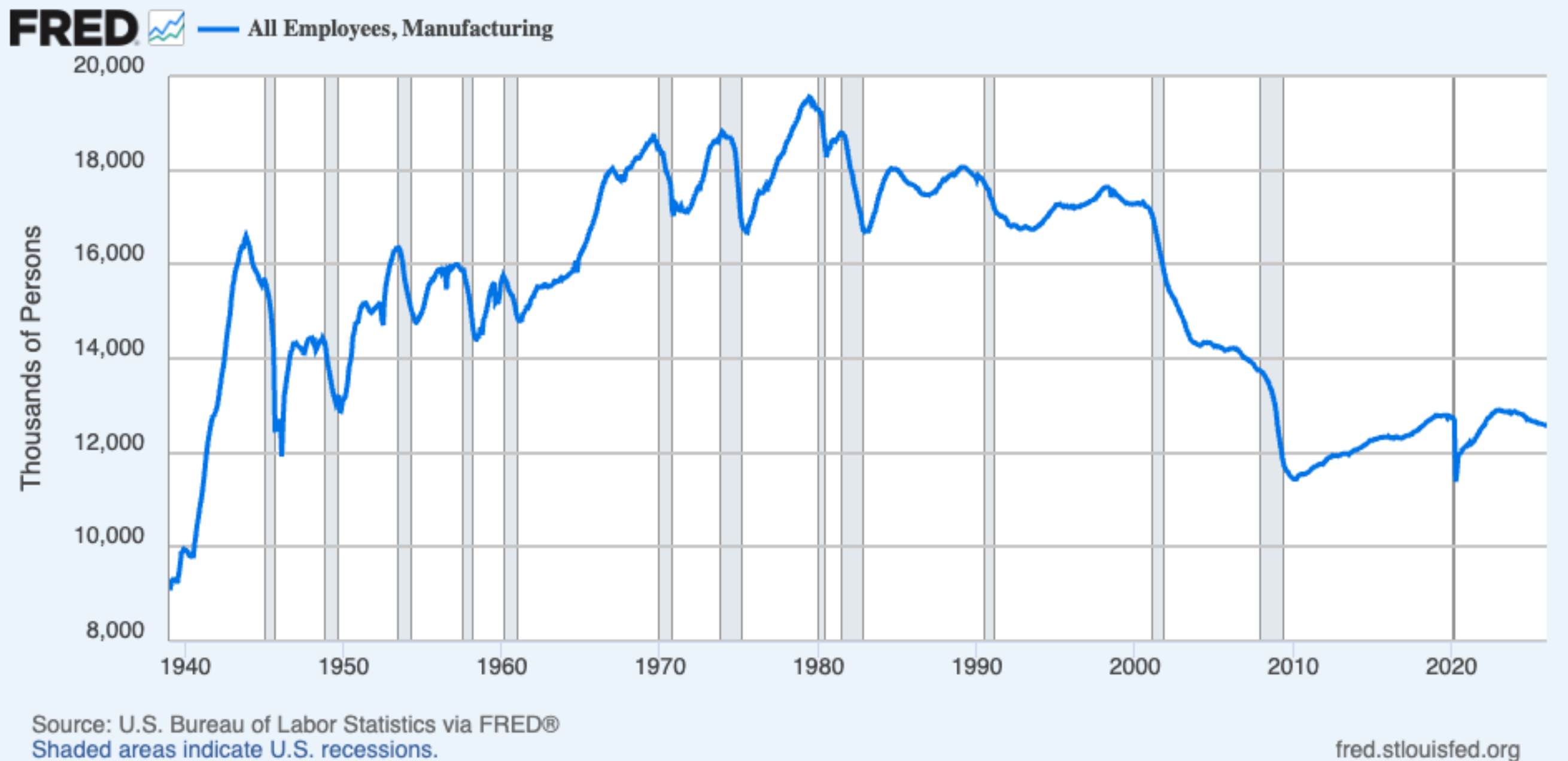


Source: Daron Acemoglu and Carlos Molina, private communication, April 2, 2023. The college wage premium is computed after controlling for various demographic characteristics such as region, age, race, gender, and experience.

Big declines in U.S. manufacturing employment after the recession of 2001 and the entry of China into the World Trade Organization in December 2001

(The wild thing is that manufacturing jobs might not pay that much anymore; [Bayard et al. FEDS 2022-011](#))

More in ECON 151. But we'll also talk about trade later in ECON 100B



Source: <https://fred.stlouisfed.org/series/manemp>